



SiamIST: Infrared small target tracking based on an improved SiamRPN

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ABSTRACT

The continuous tracking of infrared dim-small targets is significant, due to limited spatial resolution and low thermal features. Tracking algorithms based correlation filter may perform not well referring to infrared information. Therefore, a Deep Learning (DL) model is proposed for the tracking task with public data sets of small targets. To be specific, the Siamese Region Proposal Network (SiamRPN) is improved by the style recalibration module, which can obtain the perception of image styles. Furthermore, the proposed algorithm takes advantage of transfer learning technology referring to labeled target images, obtaining good features. To distinguish the small target from the background edges, the side window filtering is combined with the improved SiamRPN model. The experimental results show the good performance of the proposed small target tracking, namely SiamIST, in public near-infrared videos, compared to several related algorithms. Importantly, the designed algorithm uses the DL model to track small infrared targets for the first time, achieving an overall precision of 78.8%.

1. Introduction

Infrared Radiation (IR) information is helpful for the detection and tracking of small targets. Continuously estimating the state of a target is uneasy with its small size, complex background, and false targets with a similar representation. Moreover, small target tracking specifically focuses on detecting objects that are relatively maneuverable or low in the thermal feature. Overcoming these challenges requires the development of algorithms that can effectively distinguish small targets in a complex environment [1].

Traditional small target tracking algorithms contain detect before the track methods [2,3] and track before detect methods [4–6]. The former method enhances the target's information with the spatio-temporal filter. However, this method cannot effectively follow the false target. The latter method determines the trajectory of the target using its energy accumulation [6]. However, this method has high complexity. With a strong background of clutter and weak targets, these two types of algorithms cannot obtain accurate tracking results [7,8].

Recently, both Discriminative Correlation Filter (DCF) [1,9–17] and Deep Learning (DL) are popular in the field of object tracking. The DCF method obtains a correlation filter that identifies the object and its neighborhood [9]. The first DCF model of minimum output sum of squared error [10] utilizes the gray feature, but it fails to track the target with change in size. The Kernelized Correlation Filter (KCF) [11] does well in extracting features, and obtains a high speed benefiting from the kernel trick. However, the performance of the model is

sensitive to the boundary effect. Furthermore, the spatially regularized discriminative correlation filters [12] and the Spatial-Temporal Regularized Correlation Filters (STRCF) [13,14] add regular terms, which prevent the corruption of models. To adapt to small IR target tracking, Qian et al. [18] combine background suppression of singular value decomposition with the DCF model. The manual parameter makes this algorithm sensitive to a complex background. In addition, Qian et al. [18] take advantage of guided image filtering and KCF tracker. However, it should be noted that the guided filter may not be effective in handling noise patches that share similar representation with the target patch. In short, these DCF methods effectively address some challenges in object tracking, but little information makes it difficult to track the IR target [16,17]. There are other methods based on the sparse representation [19] and the particle filter [20]. However, real-time performance is also an important factor in the field of object tracking.

Deep learning models, such as the deep learning tracker [21], multi-cue correlation based tracker [15], Convolutional Neural Network (CNN) [22], and the Siamese Region Proposal Network (SiamRPN) algorithm [23,24], can obtain features with strong classification performance to some certain. In fact, it is unrealistic for these DL models to track small targets without training images. In recent years, some labeled IR data sets [15,25–28] referring to detection and tracking algorithms have been publicly available, providing the possibility for DL models to track small targets.

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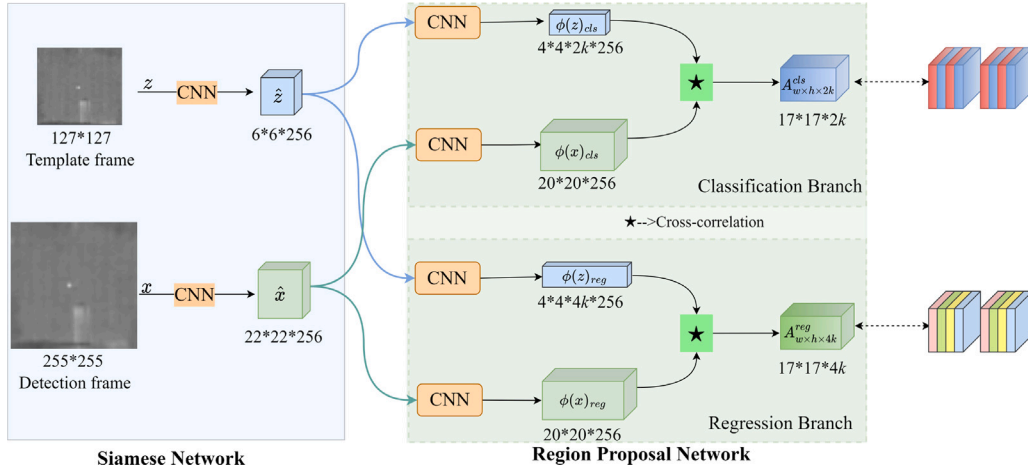


Fig. 1. The framework of SiamRPN.

According to the above analysis, a DL based algorithm is proposed for tracking small targets in Near-Infrared Radiation (NIR) videos. With Transfer Learning (TL) technology that uses a public data set of the small target [29], the proposed DL model is designed to adapt to the change in the representation of the targets and the complex background. With SiamRPN as the backbone, an effective model that utilizes Style Recalibration Module (SRM) and Polynomial Features (PF) are integrated, in order to improve the capability of feature extraction. Furthermore, the Side Window Filter (SWF) is adopted to suppress the influence of the edge edges of the image. The experimental results indicate that the proposed tracking algorithm can achieve accurate tracking results in a complex background. The main contributions are summarized below.

- A deep learning-based algorithm is proposed for the first time to track a small maneuvering target of Unmanned Aerial Vehicle (UAV) on NIR videos.
- An improved SiamRPN model utilizing SRM-PF is designed to effectively complete small target tracking.
- The SWF is adopted to suppress the bad influence of the image edges, helping to obtain a good response for small targets.
- Transfer learning is employed on a public data set of the IR small target.

The main contents of this paper are described below. Section 1 gives the introduction. Section 2 contains the details of the migration analysis and the proposed tracking algorithm. Section 3 gives the experimental result and analysis. Finally, the conclusion is drawn in Section 4.

2. The proposed SiamRPN based tracking algorithm for small objects

2.1. SiamRPN

The SiamRPN algorithm relies on pretrained networks to transform the tracking task into a one-shot detection. As shown in Fig. 1, this model contains two components, the feature extraction model in the Siamese network [30,31], and the Regional Proposal Network (RPN) [32,33] responsible for generating candidate regions. Moreover, the main task is to extract the information of a target from template frames, and infer it into detection frames, establishing a mutual similarity association.

The processed template frame is used to obtain the weights of the model. Then, the detection frame is processed on the basis of the

target's state in the previous frame. Further, the above frames are cross-correlated with each other, which can be represented by

$$A_{w \times h \times 2k}^{cls} = \phi(x)_{cls} * \phi(z)_{cls} \quad (1)$$

$$A_{w \times h \times 4k}^{reg} = \phi(x)_{reg} * \phi(z)_{reg} \quad (2)$$

where $*$ represents the operation of cross-correlation. $A_{w \times h \times 2k}^{cls}$ and $A_{w \times h \times 4k}^{reg}$ represent feature maps with different channel sizes, and k corresponds to the number of anchors. $\phi(x)_{reg}$ and $\phi(x)_{cls}$ are the corresponding feature maps referring to the search region x . In addition, template feature maps $\phi(z)_{cls}$ and $\phi(z)_{reg}$ denote convolution kernels.

In addition, the optimal loss L_{total} referring to classification and regression is described as

$$L_{total} = L_{cls} + \lambda L_{reg} \quad (3)$$

where λ (5 in experiments) is a hyper parameter used to balance the entire loss. L_{cls} denotes the cross-entropy loss, and the regression loss L_{reg} is formulated as

$$L_{reg} = \sum_{i=1}^4 L_1(\delta[i], \gamma) \quad (4)$$

where γ is utilized to control the range of the error. $\delta[1] = (T_1 - A_1)/A_3$, $\delta[2] = (T_2 - A_2)/A_4$, $\delta[3] = \ln(T_3/A_3)$, $\delta[4] = \ln(T_4/A_4)$, $\ln$ means logarithmic function. Specifically, (A_1, A_2) denotes the center point of an anchor and (A_3, A_4) denotes the corresponding shape. Furthermore, (T_1, T_2, T_3, T_4) represents the ground truth of the target. In addition, the smooth loss of L_1 is described as

$$L_1(u, \gamma) = \begin{cases} \gamma^2 u^2 / 2 & \text{if } |u| < 1/\gamma^2 \\ |u| - 1/(2\gamma^2) & \text{if } |u| \geq 1/\gamma^2 \end{cases} \quad (5)$$

2.2. Migration analysis

According to the above description, the SiamRPN method focuses on large targets on visual videos. However, the dim-small target makes this method difficult to obtain effective features. In addition, it is a challenge for classical tracking methods without training to track small targets under a complex background. Moreover, the model can mistakenly mark noise as targets, reducing the tracking performance. The development of efficient denoising methods, which can also enhance the visibility of targets, is required. In addition, insufficient generalization of training data sets needs to be compensated by the model's nonlinearity. Therefore, it is urgent to design an enhanced non-linear model to achieve style transitions.

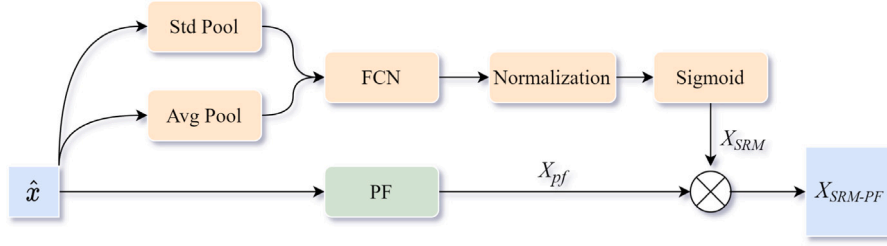


Fig. 2. SRM with Polynomial Features.

2.3. The proposed small object tracking

2.3.1. The SRM and polynomial features

Visual models struggle to obtain good IR features and are difficult to transition between RGB and IR information styles. In addition, the DL model can be easily confused by the complex background. Therefore, the Style Recalibration Module (SRM), which comes from the field of style transfer [34], is introduced to improve the performance of the SiamRPN.

The SRM improves the perception of different features by learning the correlation between style features. It can be divided into style pooling and style integration. Style pooling includes the Global Average Pooling (Avg) P_{avg} that can capture global information from the entire feature map, and the Standard (Std) deviation pooling P_{std} that can enhance the perception ability of the network. For style integration, it enhances the diversity of features by fusing the global style features and salient features. The output $X_{SRM} \in \mathbb{R}^{C \times H \times W}$ of SRM can be represented by

$$X_{SRM} = \sigma_{BN} \{ f^{2 \times 2} [P_{avg}(\hat{x}) + P_{std}(\hat{x})] \} \otimes \hat{x} \quad (6)$$

where $\otimes$ represents the matrix multiplication, σ_{BN} denotes the operation of batch normalization and sigmoid function, and $f^{2 \times 2}$ denotes a 2×2 operation of full convolutional network. The feature image $\hat{x} \in \mathbb{R}^{C \times H \times W}$ denotes an output of the Siamese network (Fig. 1), where C , H , and W represent the corresponding numbers of channels, height, and width, respectively.

During training, limited data can make the model converge quickly. Several factors, such as insufficient training samples and poor data quality, make the model unable to effectively learn the correlated information between the template frame and the subsequent frame. Also, the complexity of the model can affect the performance of the SRM model. Therefore, polynomial features, which can enhance the non-linearity of the DL model, are introduced in this paper. Specifically, the PF X_{pf} can be described as

$$X_{pf} \approx a \times \hat{x} + b \times \hat{x}^2 \quad (7)$$

where parameters a , b , and c need to be appropriately selected (Only utilizing quadratic terms in experiments, i.e., $c = 0$). On the basis of this, the feature X_{SRM} can be updated to X_{SRM-PF} .

$$X_{SRM-PF} = \sigma_{BN} \{ f^{2 \times 2} [P_{avg}(\hat{x}) + P_{std}(\hat{x})] \} \otimes X_{pf} \quad (8)$$

Subsequently, the combination of PF and SRM is shown in Fig. 2. The ablation experiment is conducted in Section 3.4.

2.3.2. Side window filtering

The above SRM-PF enhances style perception and feature representation. However, tracking a small object that is partially occluded by the edges of the image remains a difficult problem. Thus, side window filtering is adopted to suppress the bad influence of edges. Specifically, the SWF enumerates eight directions (Up, down, left, right, northeast,

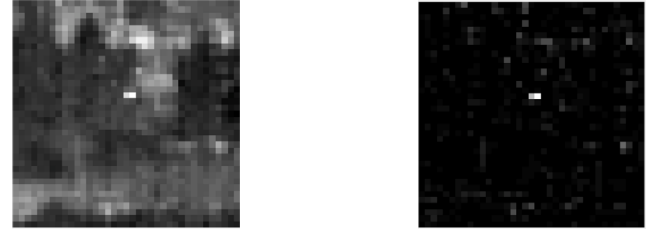


Fig. 3. The effect of SWF referring to small target.

southeast, southwest, and northwest) for the filtered window, referring to edges and corners. Given a region x of interest, the filtered result $x^{r,\rho,\theta}$ of SWF can be represented as

$$x^{r,\rho,\theta} = G(x, r, \rho, \theta) \quad (9)$$

where G means the SWF function, and the radius of the filtered window is r . $\rho = 0$ or r indicates that the pixel is located at the edge and corner of the window, respectively. In addition, $\theta = 0, \pi/2, \pi, 3\pi/2$ represents four directions of the filtered window.

Furthermore, the optimal result $S(x)$ is obtained by minimizing the distance between the input x and the output $x^{r,\rho,\theta}$.

$$S(x) = \arg \min_{\forall x^{r,\rho,\theta}} \|x - x^{r,\rho,\theta}\|_2^2 \quad (10)$$

where $\|\cdot\|_2$ denotes the 2-norm, and the target's information X is enhanced by $(x - S(x))$.

Subsequently, the edge-preserving capability of the SWF model is proved on infrared images. In SWF, each radius of the filter window is set to 1, and the number of iterations is 3. The experimental result is shown in Fig. 3(a). It is evident that the complex background can be suppressed, and the target retains its high-frequency information. In other words, the information of targets and edges can be effectively distinguished, which is a challenge for traditional methods that use box filtering technology.

2.3.3. Transfer learning

In small target tracking, the features extracted by the visual DL model are not appropriate enough for effective target matching. Therefore, transfer learning [35–37] is adopted to enhance the generalization performance of the DL model. Specifically, the TL model aims to learn generalized feature representations and model parameters from the source domain D_s with large-scale data and apply them to the target domain D_t with limited data. In detail, the feature extraction model is pre-trained using RGB videos, and the entire model is fine-tuned with infrared videos, which is shown in Fig. 4.

As a fact, the spatial resolution of infrared images is low, that is to say, it is incapable for the feature extraction model to capture sufficient

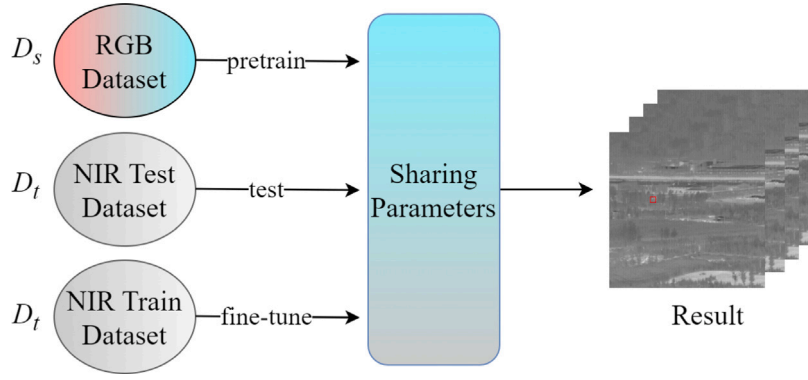


Fig. 4. The schematic diagram of transfer learning.

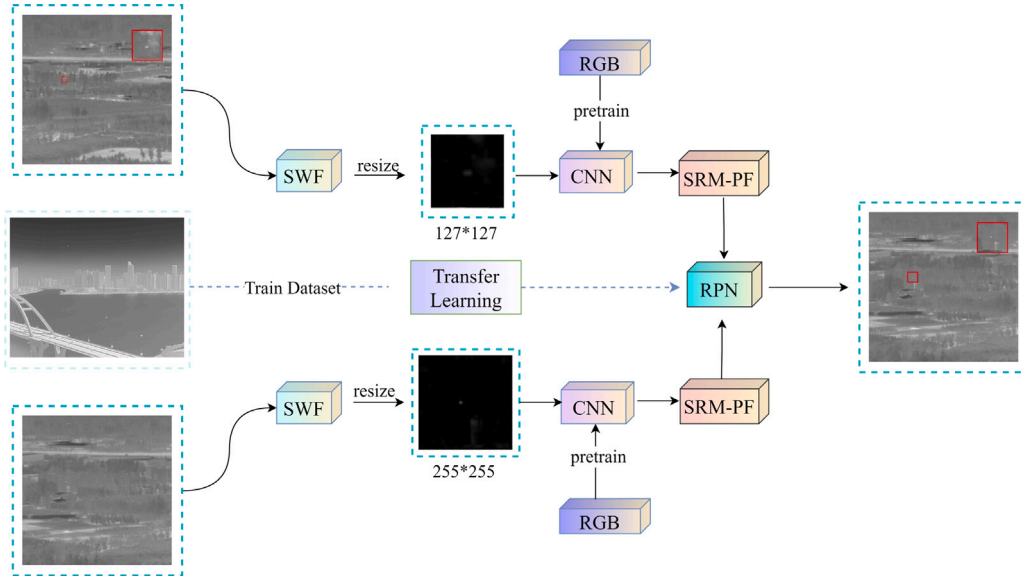


Fig. 5. The flowchart of the proposed SiamIST algorithm.

detail for IR target tracking. In summary, these designed techniques, including the SRM-PF model in feature representations, and the SWF model in the improved SiamRPN model, play key roles in the task of small target tracking.

2.3.4. The flow of the proposed algorithm

The proposed algorithm consists of two parts, a preprocessing stage in which the image is processed with SWF, and an improved SiamRPN that combines SRM with non-linear PF. The algorithm step is described as follows, and the corresponding flow diagram is shown in Fig. 5.

1. Preprocess the template frame utilizing SWF.

- The position of the target in the initial frame is initialized, obtaining the sampling region z .
- The feature of the small target Z is obtained by subtracting the filtered result $S(z)$ of SWF from the template frame z .

$$Z = z - S(z) \quad (11)$$

2. The feature Z_{SRM-PF} obtained in Z is further processed by SRM-PF for style perception and model's non-linearity, according to Eq. (12).

$$Z_{SRM-PF} = \sigma_{BN} \{ f^{2 \times 2} [P_{avg}(\hat{Z}) + P_{std}(\hat{Z})] \} \otimes (a \times \hat{Z} + b \times \hat{Z}^2) \quad (12)$$

where $\hat{Z}$ refers to CNN operations of Z .

- Two distinct convolutional kernels are generated from the feature Z_{SRM-PF} for the RPN model.
- The detection frame x is processed with the same operation as the template frame.

$$X_{SRM-PF} = \sigma_{BN} \{ f^{2 \times 2} [P_{avg}(\hat{X}) + P_{std}(\hat{X})] \} \otimes (a \times \hat{X} + b \times \hat{X}^2) \quad (13)$$

where $\hat{X}$ is the corresponding CNN operation of $X = x - S(x)$, which denotes the small target feature with SWF.

- The feature X_{SRM-PF} of the detection branch continues to undergo the operation described in step 3.
- With step 3 and step 5, the cross-correlation between the classification and regression branches is established as Eqs. (1) and (2), and the position of the target is updated.
- Go to step 4, complete the continuous tracking of objects.

3. Experiments and analysis

The experimental setup is described in Section 3.1, including the selected data set and experimental details. In Sections 3.2 and 3.3, the qualitative and quantitative analyzes are performed with several existing small object tracking methods. The strengths and limitations of the proposed algorithm are discussed. Furthermore, in Section 3.4, ablation experiments are performed to demonstrate the effectiveness of each component in tracking small objects in NIR videos.

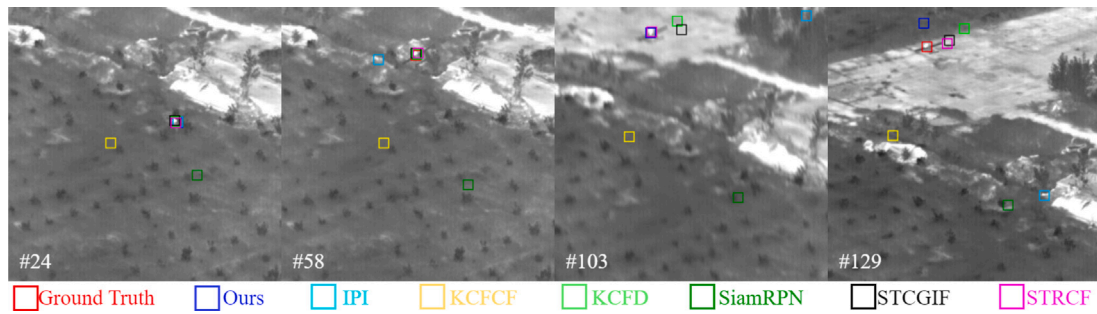


Fig. 6. Qualitative results over Video 9.

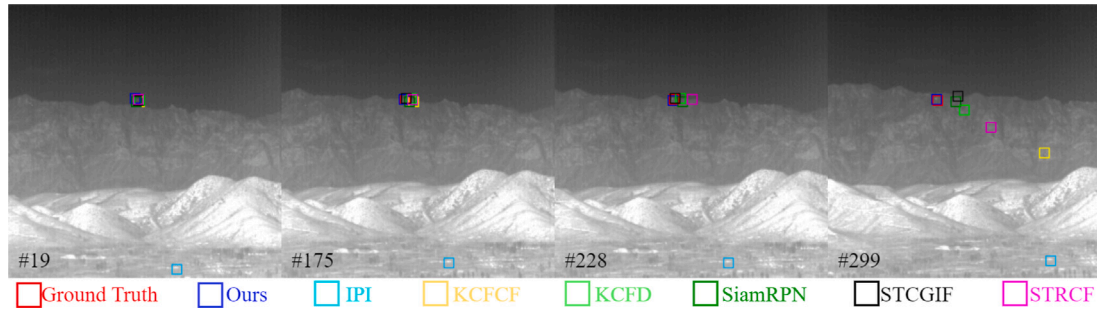


Fig. 7. Qualitative results over Video 10.

3.1. Experiment setup

On the purpose of evaluating the performance of the model, the small target datasets of public IR [29,38] are used for training and testing. The training data set¹ derives from ‘A dataset for small infrared moving target detection under clutter background’ [29], comprising diverse scenes with complex background, shooting angles, and motion scenarios. Selected videos primarily contain targets of 3×3 pixels, meeting the SPIE’s definition of small targets.

In addition, the test data set² is obtained from ‘A dataset for infrared image dim-small aircraft target detection and tracking under ground/air background’ [38]. This data set includes real UAVs under complex background, such as sky and ground. The first four videos are excluded because the target is oversized or multiple.

For dim-small targets, the size of the tracking boxes is represented by 9×9 pixels. Considering the testing efficiency and the target scale, the window radius in SWF is set to 1. In SRM-PF, both a and b are set as 1. In SiamRPN, the aspect ratio of the anchors is also manually set to 1. Comparison algorithms contain Infrared Patch Image (IPI) [39], KCFCF [18], KCFD [40], SiamRPN [23], STCGIF [41], and STRCF [13]. Experiments are conducted using PyCharm on a computer with a 3.8 GHz CPU and 32 GB RAM. Codes can be requested by email.³

3.2. Qualitative comparison

Experimental results on six representative IR videos, including videos 9, 10, 15, 16, 17 and 22, are selected for demonstration, as shown in Figs. 6 to 11.

In Fig. 6, the target on video 9 undergoes a complex background of vegetation, and there is a sudden increase in the grayscale characteristic. The IPI model mistakenly assumes similar thermal objects as

targets. The SiamRPN model without TL and SWF loses the target over the entire video. Further, only the proposed algorithm performs well at frame #103, overcoming the difficulty of strong edges. However, all algorithms fail to track the small UAV at frame #129, for the reason of that the target moves fast under strong background.

In Fig. 7, the dim target is located among the mountain and the sky. Specifically, both the KCFCF and the STCGIF drift to the background edge in frame #175, and cannot be detected again in frames #228 and #299. KCFD performs well before frame #228. Also, only the proposed Siamese model, which does well in extracting small target characteristics, successfully obtains the position of the target in the frame #299.

In Fig. 8, there appears prominent point regions with similar features to the region of the target. Furthermore, there is a slight movement in frame #59, resulting in failure of all algorithms. All DCF based algorithms (KCFCF, KCFD, STCGIF and STRCF) tend to detect the building as a target. However, the proposed algorithm successfully re-tracks the small target in frame #88 in a large search region.

In Fig. 9, the resolution of video 16 is low, and the rapid movement of the target results in a blurry image, as observed in frame #214. KCFD cannot find the target after frame #214, which indicates that the detection method in KCFD is not effective. More to the point, the original Siamese model loses the target quickly due to a bad feature extraction model. When considering the distinction of targets from edges, the improved Siamese algorithm can track the target effectively.

In Fig. 10, strong edges affect the performance of the trackers. The IPI algorithm obtains a false target from beginning to end. The STRCF model remains in regions with high-frequency characteristics after frame #200. All other algorithms perform relatively well, due to the fact that the background is simple. However, the SiamRPN algorithm drifts to the background, which means that this model is unstable when tracking small targets.

As shown in Fig. 11, the background is complex and there are a large number of trees around the UAV target. In detail, the tracking error referring to KCF based algorithms is large at most of frames. Both two Siamese-based algorithms obtain good tracking results, because of the fact that the feature of the target is relatively salient, and there is a great difference between the patch of a target and its local region.

¹ Available at: <https://www.scidb.cn/en/detail?dataSetId=808025946870251520>.

² Available at: <https://www.scidb.cn/detail?dataSetId=720626420933459968>.

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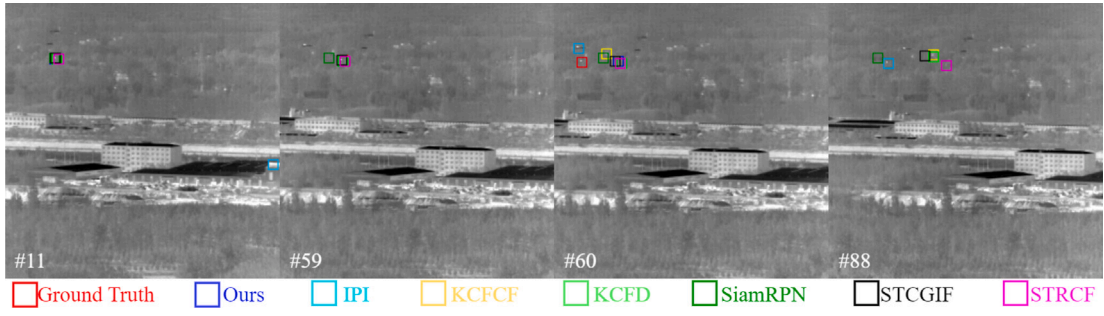


Fig. 8. Qualitative results over Video 15.

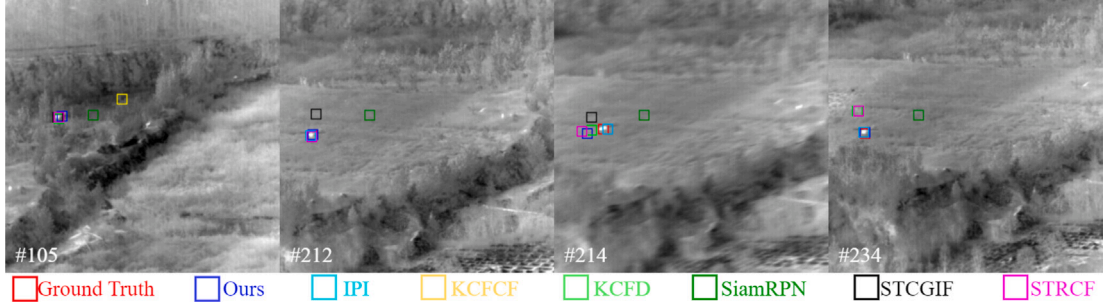


Fig. 9. Qualitative results over Video 16.

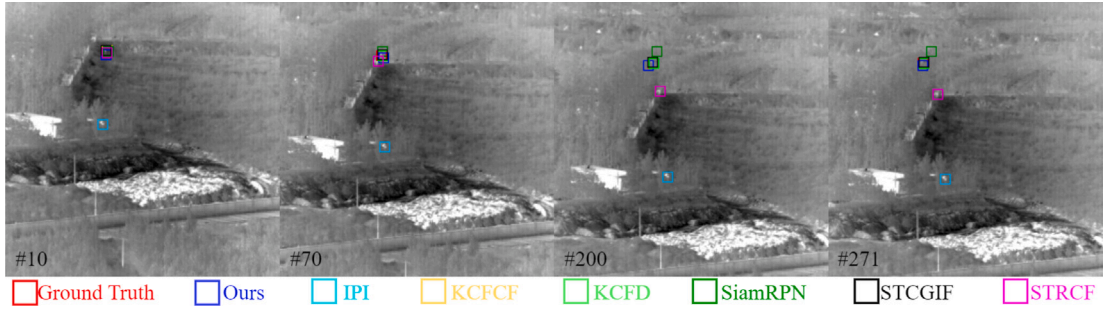


Fig. 10. Qualitative results over Video 17.

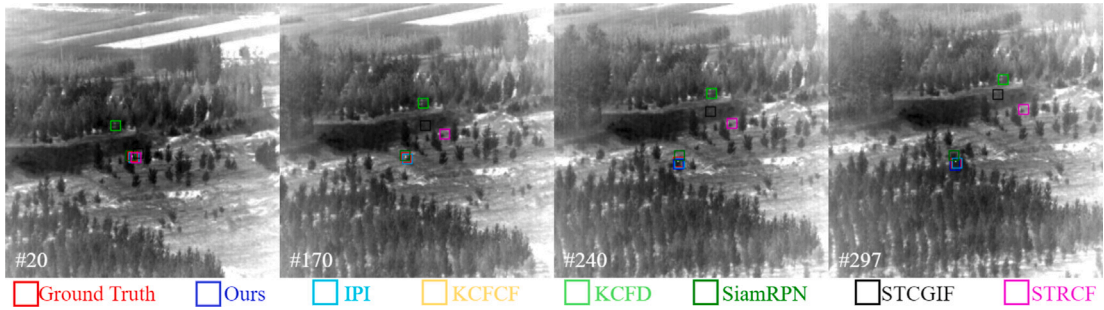


Fig. 11. Qualitative results over Video 22.

3.3. Quantitative comparison

In this section, performance is evaluated using precision plot and success plot [42]. Precision can be described as the ratio of frames, in which the Center Location Error (CLE) between the predicted box and ground truth is below a certain threshold to the total number of frames. The success rate, also known as the overlap rate, represents the ratio of frames in which the intersection over union between the predicted bounding box and ground truth is above a certain threshold to the total

number of frames. In addition, the Area Under the Curve (AUC) of the success curve represents the area under the performance curve with respect to the axis of abscissas.

As shown in Fig. 12, all related algorithms are tested on the NIR dataset, in which each video contains 300 frames. Obviously, the proposed algorithm is superior to the SiamRPN without TL in both the precision (Increase by 33%) and overlap rate (Increase by 67%). Besides, the KCFCF and the KCFD obtain good performance, benefiting from the image filtering technology.

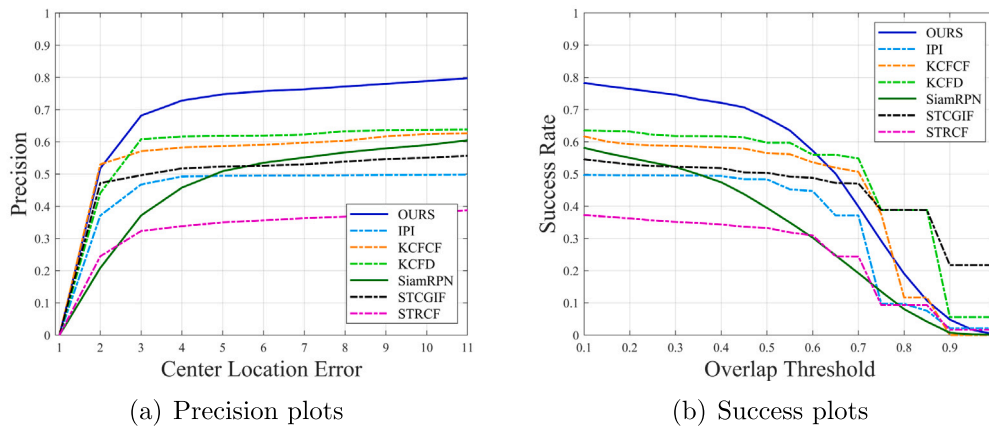


Fig. 12. The performance comparison.

Table 1

Performance comparison of algorithms (Precision referring to CLE = 10 and FPS).

Algorithm	AUC	Overall precision (CLE = 10)	Overall FPS
IPI [39]	0.371	0.497	1
KCFCF [18]	0.465	0.624	69
KCFD [40]	0.509	0.633	70
STCGIF [41]	0.472	0.550	50
STRCF [13]	0.281	0.378	145
SiamRPN [23]	0.311	0.590	62
Ours	0.520	0.788	30

Table 2

The ablation experiment referring to SWF and SRM-PF (CLE equals to 10).

Algorithm	AUC	Precision (CLE = 10)	Overall FPS
Baseline	0.311	0.590	62
Baseline-SWF	0.321	0.671	30
Baseline-SWF-SRM	0.320	0.670	30
Baseline-SWF-SRM-PF	0.520	0.788	30

SWF denotes Side Window Filtering, SRM denotes Style Recalibration Module, and PF denotes Polynomial Features.

Furthermore, Table 1 shows the performance comparison, which includes the precision, the AUC of success rate curves, and the Frame Per Second (FPS), referring to all NIR videos. It can be observed that the KCFD exhibits accurate tracking of small targets, with the second highest value of AUC (0.509). Moreover, the FPS of the STRCF ups to 145, but this model performs not well in tracking small target, largely because of strong edges. Besides, the baseline algorithm achieves nearly two times faster than the proposed algorithm in terms of FPS, for the reason of SWF. Yet, the proposed algorithm excels in overall precision (0.788) and AUC (0.52) with CLE equal to 10.

In short, the proposed algorithm performs well, as evidenced by the visual effect and high evaluation value.

3.4. The ablation study

In order to validate the effectiveness of the proposed algorithm compared to baseline, tests are carried out on SiamRPN with the addition of SWF, SRM, and SRM-PF to the whole NIR dataset. Table 2 demonstrates that the baseline achieved a reasonable precision of 0.59, indicating a moderate level. However, Table 1 reveals that the corresponding AUC for the success rate of the baseline is relatively low.

As shown in Table 2, both the SRM-PF and the SWF help to improve the tracking performance. On the one hand, the baseline-SWF achieves the overall precision of 0.671, which is better than that of the baseline algorithm (0.59). On the other hand, the baseline-SWF-SRM-PF scores the highest precision value (0.788), proving the effectiveness of the SRM-PF model.

4. Conclusion

An improved SiamRPN algorithm is designed in this paper to track a dim and small target on NIR videos. The non-linear SRM-PF adapts to the IR information by fine-tuning weights, enhancing the performance of feature extraction. The high quality of the training set in the public data set plays a key role for the generalization of our tracker. Furthermore, the TL technology makes the SiamRPN-based algorithm perform well in extracting the feature referring to small targets. Additionally, a suitable radius in the SWF helps suppress the bad influence of background edges. Most importantly, experimental results insist that the DL based algorithm of IR small target tracking achieves the overall precision of 0.788. This lays the foundation for applying deep learning to track small targets.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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